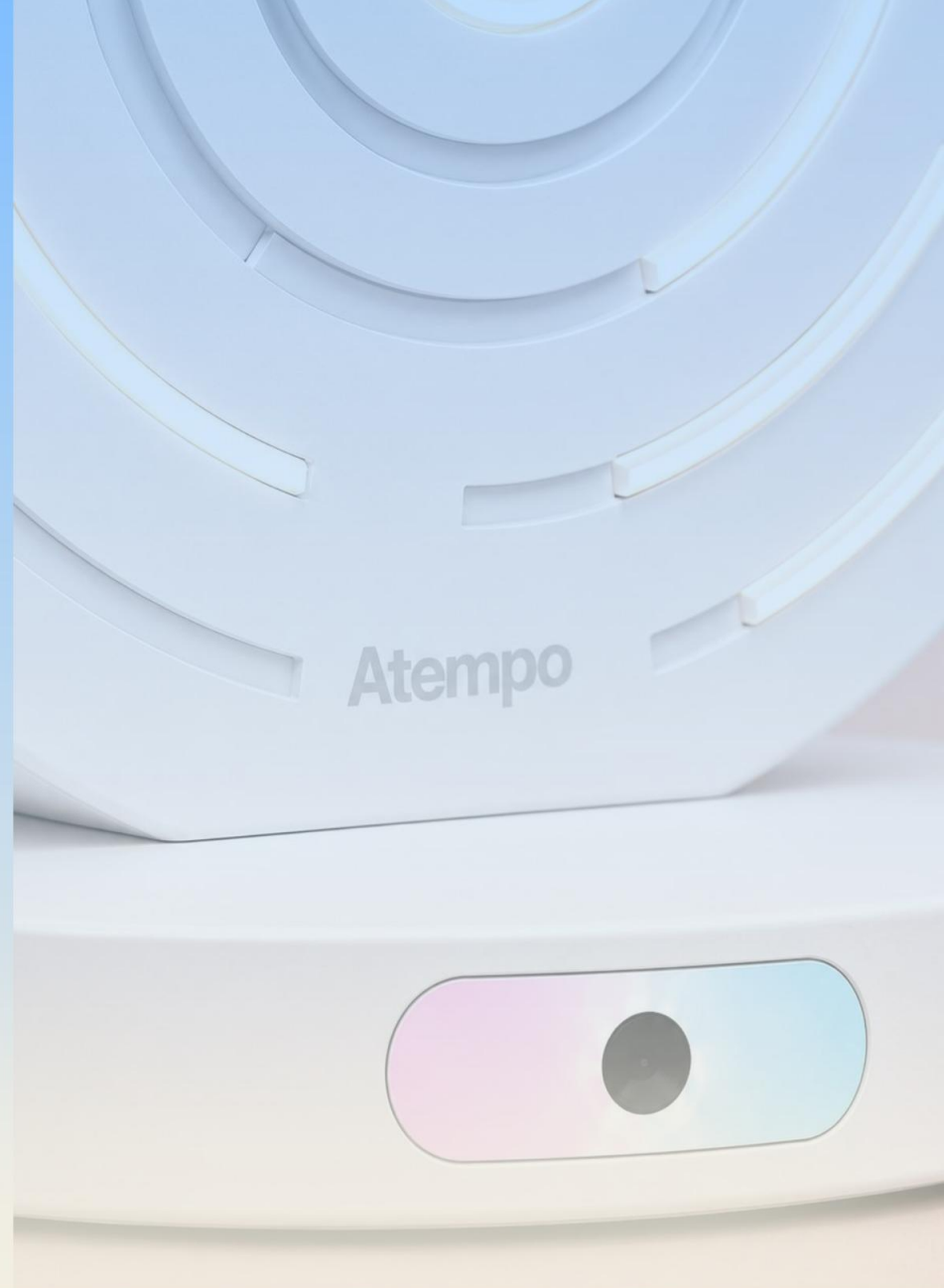


Atempo 缓律

基于非接触呼吸感知与环境式生物反馈的桌面节律交互系统

A Desktop Interaction System for Rhythm Regulation through
Contactless Respiration Sensing and Ambient Biofeedback



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01 研究背景

1

问题切入

Task ends,
but recovery may not begin.

高认知负荷任务结束后，人并不会自动进入恢复状态。

在“恢复”研究中，**心理抽离（psychological detachment）** 被视为关键体验：

个体需要从工作/学习相关的思考、紧张和任务惯性中切换出来，恢复才真正开始。

这使“任务结束后的几分钟”成为一个可被设计介入的过渡窗口。

Shimazu, Akihito, et al. "Psychological Detachment from Work during Non-Work Time: Linear or Curvilinear Relations with Mental Health and Work Engagement?" *Industrial Health*, vol. 54, no. 3, 2016, pp. 282-292. <https://doi.org/10.2486/indhealth.2015-0097>

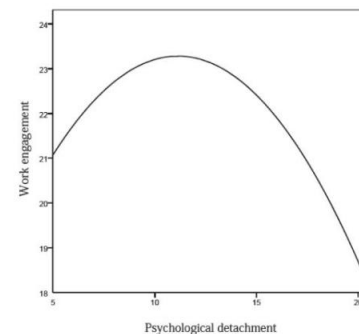


Fig. 4. Curve-fitting between psychological detachment and work engagement.

Detachment → Work Engagement

恢复不是从任务中彻底逃离，而是在适度抽离中重新找回稳定节奏

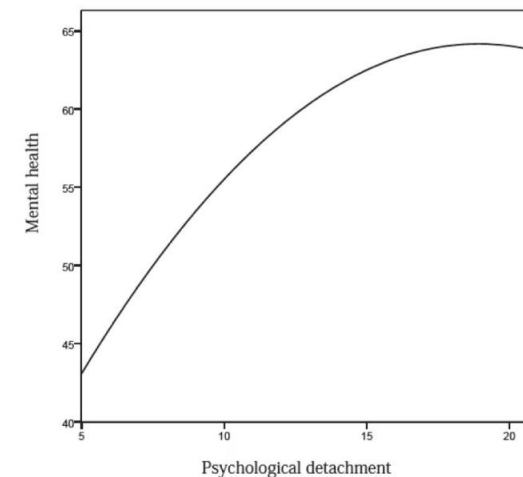


Fig. 3. Curve-fitting between psychological detachment and mental health.

Detachment → Mental Health

研究基于日本第三产业正式雇员的横断面互联网调查



2,234 participants

Psychological Detachment + Relaxation

02 设计机会

竞品矩阵分析



2

介入窗口

现有产品大多要求用户“穿戴进入”或“主动训练进入”；
而任务结束后的桌面充电停留，本身就是一个天然但尚未被充分设计的状态切换窗口。

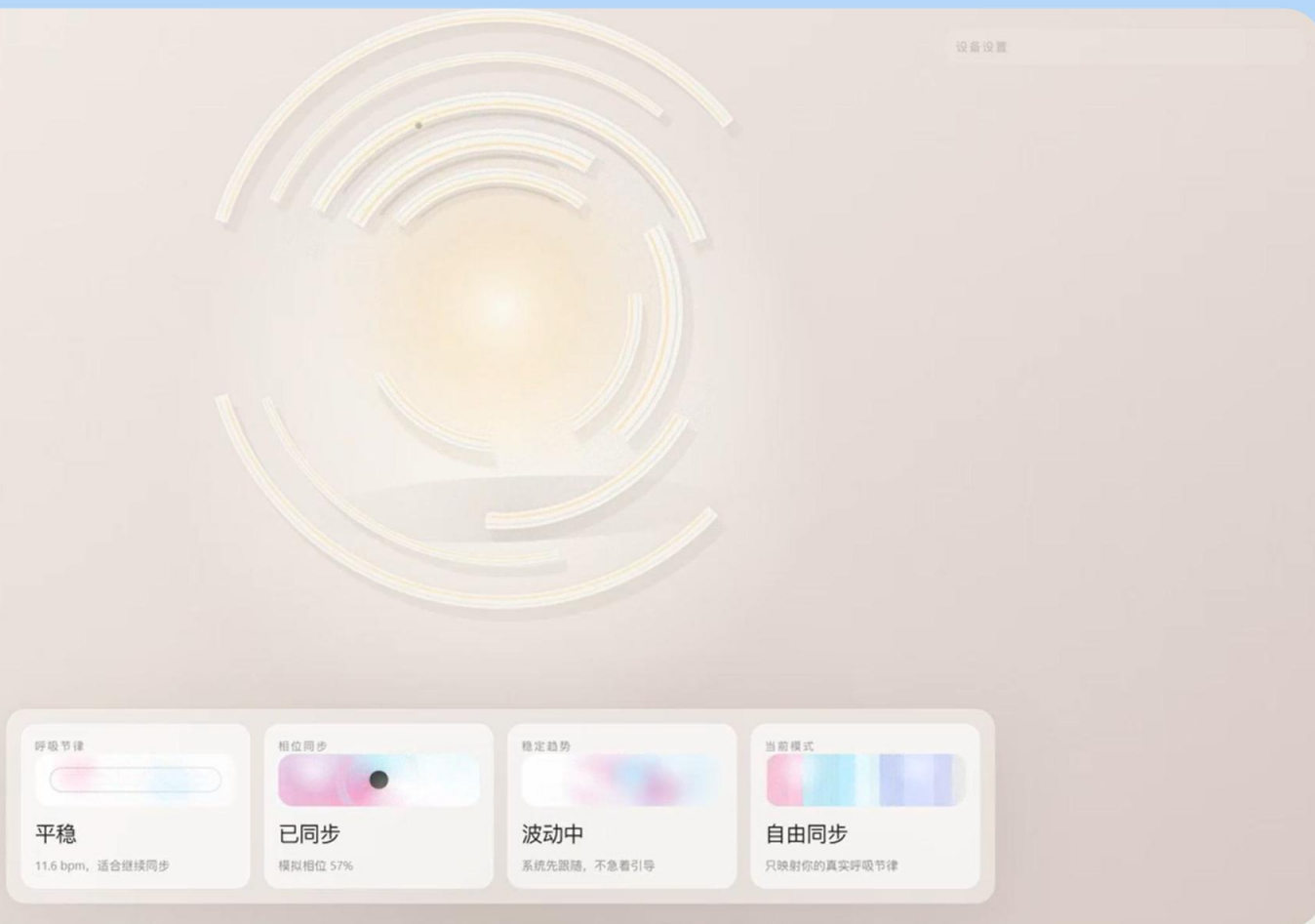
Atempo 选择切入这个窗口，没有试图判断用户处在“焦虑”还是“放松”的标签里把 biofeedback 从“长期监测”与“干预训练”之间，转化为一种更低门槛的桌面恢复流程：

1. 充电 提供自然停留时间
2. 设备归位 提供进入仪式
3. 双弧靠近 提供节律反馈
4. 稳定光场 提供回稳确认

在用户已经会发生的桌面行为里，嵌入一次可被完成的状态收束。

产品定义

Atempo 不是情绪识别设备，而是一个将充电停留转化为呼吸回稳过程的桌面节律交互系统。



03 交互机制

3

场景定义

充电停留 → 呼吸显影 → 双弧靠近 →
节律同步 → 回稳陪伴

目标用户

大学生 / 青年上班族

典型情景

结束高强度学习、工作或创作后，用户回到书桌或私人空间，身体虽已离开任务现场，但注意力仍停留在任务惯性中。此时，充电成为一个自然的停留动作。

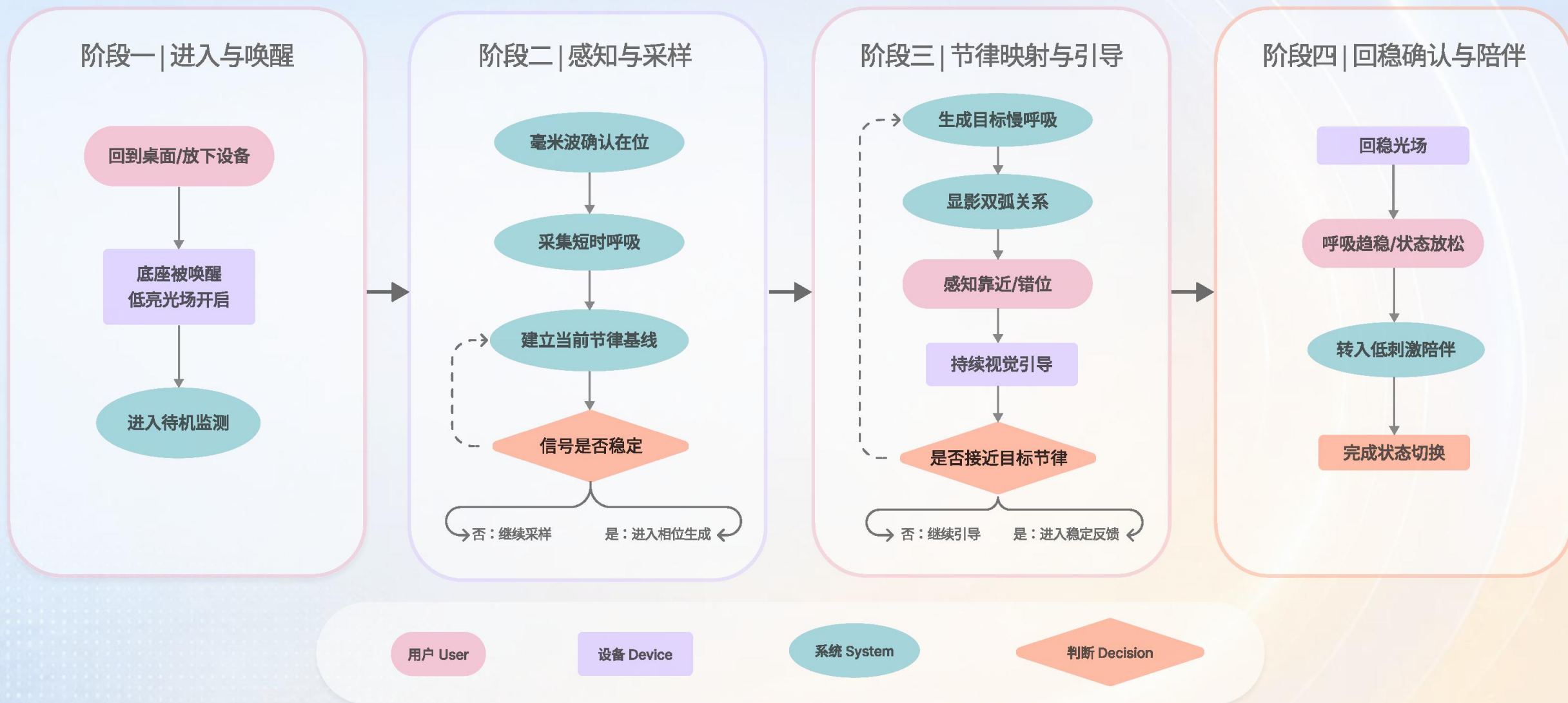
Scenario Framing



核心目标

通过“充电停留—呼吸显影—节律靠近—回稳陪伴”的连续交互，帮助用户看见当前呼吸与目标慢呼吸之间的关系，并在跟随中逐步回到稳定节律。

3 交互流程



3

呼吸引导与生理反馈的理论依据

呼吸是调节自主神经系统的入口

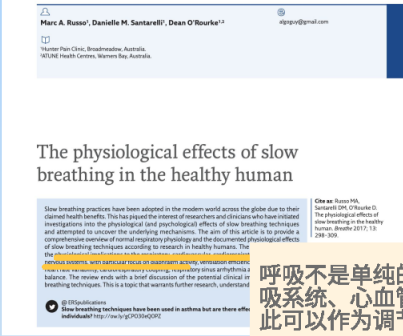
呼吸可以被视为进入自主神经系统调节的低门槛入口。相比直接控制心率或情绪，用户更容易主动控制呼吸节奏，并通过呼吸影响身体的紧张、恢复与稳定状态。

0.1Hz帮助身体进入稳定节律

resonance frequency breathing 的理论，将 **0.1 Hz，即约每分钟 6 次呼吸** 作为引导的目标节奏会带来明显的心血管节律变化

中度复杂度带来舒适的视觉吸引力

Robles et al. 的分形设计研究指出，分形偏好来自视觉吸引力与放松感之间的平衡，视觉复杂度过低会单调，过高会刺激，**中等至中高复杂度**更容易兼顾吸引力与恢复感。



呼吸不是单纯的空气交换，而是会影响呼吸系统、心血管系统和自主神经系统，因此可以作为调节身体状态的入口。

Implications

Variations in respiratory rate within the normal range of 6–10 breaths per minute have been found to be associated with health outcomes. “自主神经系统优化的呼吸”似乎位于每分钟 6–10 次呼吸的范围内。 Variations in respiratory rate within the normal population, and given the profound effect on vagal tone, what constitutes “autonomically optimised respiration”. Perhaps it is time to refine a breathing technique that optimises ventilation, gas exchange and arterial oxygenation, maximises vagal tone, maintains parasympathetic-sympathetic balance and optimises the amount of cardiorespiratory reserve that could be called upon in times of intense physical or mental stress or activity. According to the studies reviewed here, “autonomically optimised respiration” would appear to be in the band of 6–10 breaths per min, with an increased tidal volume that is achieved by diaphragmatic activation. Breathing is also a tool of optimised respiration that is achievable in

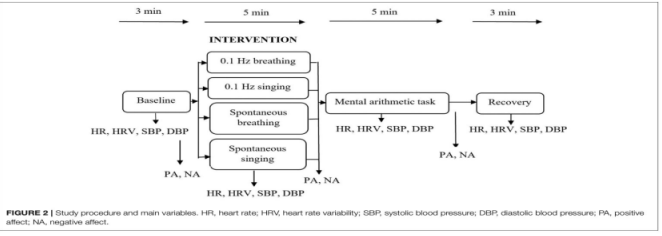
0.1 Hz，能够在多种生理系统中引发协调性的振荡。

Singing at 0.1 Hz as a Resonance Frequency Intervention to Reduce Cardiovascular Stress Reactivity?

Sandra Teasdale, Christian Rottmann, Bernhard Weber, Josef M. Tatzert and Andreas R. Schwaninger

Slow breathing at 6 breaths per min (corresponding to ~ 0.1 Hz) has been found to benefit psychological and physical health. In this study, we aimed to examine if paced singing at 0.1 Hz has beneficial acute effects on physiological reactivity compared to slow breathing. Participants were randomized to one of four experimental interventions prior to performing a mental stress task: paced breathing at 0.1 Hz (n = 26), paced singing at 0.1 Hz (n = 26), spontaneous breathing (n = 26), or spontaneous singing (n = 26). Heart rate, heart rate variability in the low (LF-HRV) and high frequency (HF-HRV) domain, blood pressure and affective well-being were assessed. As expected, both paced breathing and paced singing resulted in elevated LF-HRV. Moreover, both singing groups evidenced increases in heart rate, blood pressure and positive affect, thus indicating elevated sympathetic activation. Breathing and singing at 0.1 Hz had no robust effect on cardiovascular stress reactivity. Findings suggest that paced singing could constitute a promising alternative to slow paced breathing as it increases cardiovascular coherence, although more studies are needed to elucidate whether slow breathing and/or singing could ameliorate acute stress responses.

Keywords: cardiovascular resonance, coherent breathing, heart rate variability, mental stress, resonance breathing



Aesthetics and Psychological Effects of Fractal Based Design

Kelly E. Robles, Michelle Roberts, Catherine Veigtkens, Julia H. Smith, Oscar Rosales, Julia Rosales, Sabine Stelcher, Annette Lepp, Martin Lepp, Richard P. Taylor, Brenda Spahr and Margaret E. Sarno

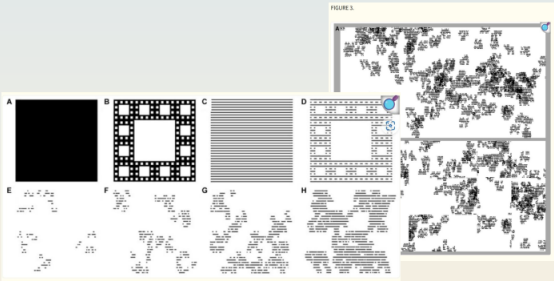
Integrated Psychological and Department of Psychology, University of Oregon, Eugene, OR, United States; *Psychology and Department of Psychology, University of Oregon, Eugene, OR, United States; †Department of Psychology, University of Oregon, Eugene, OR, United States; ‡School of Design, University of Oregon, Eugene, OR, United States; §School of Design, University of Oregon, Eugene, OR, United States

Highly prevalent in nature, fractal patterns possess self-similar components that repeat

patterns can nonrandomly patterns judgments, real forest? tal patterns at can be at patterns dist fractal

forest. The local “tree-seed” patterns, global configuration of tree-seed locations, and overall resulting “global-forest” patterns have fractal qualities. These designs span multiple mediums yet are all intended to lower occupant stress without detracting from the function and overall design of the space. In this series of studies, we first establish divergent relationships between various visual attributes, with pattern complexity, preference, and engagement ratings increasing with fractal complexity compared to ratings of refinement and relaxation which stay the same or decrease with complexity. Subsequently, we determine that the local constituent fractal (tree-seed) patterns contribute to the perception of the overall fractal design, and address

分形复杂度影响视觉体验，适度复杂更容易兼顾吸引力与放松感。



中等复杂度，让圆弧与点阵既能传达变化，又保持柔和低刺激。

3

数据转译逻辑

01 输入层 | 呼吸波动

Breath Signal

毫米波雷达感知用户胸腔微动，估测当前呼吸频率与呼吸稳定性。

02 关系层 | 目标差值

Distance to 0.1Hz

系统将当前呼吸与目标慢呼吸节律进行比较，计算二者之间的距离与靠近趋势。

03 转译层 | 双弧相位

Arc Phase Gap

当前呼吸被映射为动态弧线，目标节律被映射为稳定弧线。

两者的错位、速度与距离，表达当前节律是否正在靠近目标

04 反馈层 | 回稳确认

Stable Feedback

当呼吸节律逐渐变慢、波动减小并持续接近目标时，弧线趋于同步，光效进入稳定状态。

04 技术实现

呼吸相位钟摆系统 · 爆炸技术路径图

将人体呼吸节律转译为灯槽内的钟摆式光效与可感知的同步反馈

人体呼吸 —— 毫米波雷达 —— 相位生成 —— 钟摆反馈

软件技术路径

从数据到相位，再到钟摆运动

01 呼吸数据读取

LD6002: 存在 / 速率 / 信号强度

02 滤波与状态机

无人 / 寻找 / 跟随 / 弱信号

03 呼吸相位估计

$\text{phase} += \Delta t \times \text{breathRate} / 60$

04 钟摆运动模型

$\theta = \sin(\text{phase} \times 2\pi) \times A$

05 光效渲染引擎

粉白蓝色场 + 相位弧线 + 同步强度

硬件安装层

爆炸结构中的关键部件

前盖扩散面板

磨砂 PC / 亚克力, 柔化灯带颗粒

COB 可编程灯带

生成环形呼吸光阵与底部灯槽

物理钟摆机构

微型舵机或磁驱动, 控制摆片角度

LD6002 呼吸雷达

隐藏在面板后方, 采集呼吸节律

ESP32-S3 主控

相位计算、状态机、灯效和执行控制

5V 电源与驱动电路

为灯带、舵机/线圈和主控供电



底部相位灯槽

显示呼吸相位摆动与同步状态

核心转译公式: $\text{phase} += \Delta t \times \text{breathRate} / 60 \rightarrow \theta = \sin(\text{phase} \times 2\pi) \times A$

05 产品呈现

呼吸相位钟摆用户旅程图

从靠近设备到形成同步反馈：用户感知、产品响应与底层技术的连续路径



用户并不是在读取数据，而是在感知一个会跟随自己呼吸的情绪反馈装置。

5

外观与形式



① 桌面充电底座



② 竖向节律显示面



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THANK YOU